

Preprocessing Report — ODI Batting Dataset

Objective: Prepare an ODI batting dataset for machine-learning use through EDA, missing-value handling, encoding, scaling, feature engineering and leakage-safe splitting.

Dataset: 2,500 rows × 15 original columns.

Missing values: '-' placeholders are converted to NaN. Numeric variables use median imputation and categorical variables use most-frequent imputation.

Encoding: Team/country is extracted from Player and One-Hot Encoded. Player is dropped as a high-cardinality identifier.

Scaling: StandardScaler is fitted only on the training set inside a Scikit-learn Pipeline.

Feature engineering: Start_Year, End_Year, Career_Years, Runs_per_Innings and Team are created.

Target: High_Average is 1 for batting average at or above the median. It is included only to demonstrate the preprocessing/model pipeline.

Model-performance demonstration

Logistic Regression is used as a demonstration. The preprocessing makes the dataset consistent and model-ready, while the train-only fitting of transformations prevents test-set leakage.

Metric	Score
Accuracy	0.940
Precision	0.956
Recall	0.916
F1 Score	0.935

Outlier detection

Potential outliers were identified with the IQR rule. They are reported rather than blindly deleted because extreme cricket statistics can be legitimate high-performing observations.

Conclusion

The final cleaned dataset is exported as data/processed_dataset.csv. The notebook documents the full reproducible workflow and the visualizations show the Runs distribution before and after StandardScaler.